

NetPath Test Case Specification

High-Level Test Cases and Use Case Traceability Matrix

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1. Document Purpose

This Test Case Specification defines high-level functional test cases for the main use cases of NetPath, an interactive, browser-based networking education platform. The test cases focus on the core learner, instructor/admin, and platform functions described in the proof of concept: guided tutorial stages, drag-and-drop sandbox design, device configuration, topology validation, animated request simulation, saved topology portfolio, authentication/progress persistence, and cohort progress monitoring.

Source basis: NetPath Proof of Concept / Business Plan, March 2026.

2. Scope and Assumptions

- Testing is written at a high level and is suitable for system, integration, and user acceptance testing.
- The test environment contains seeded curriculum data, test users, and sample instructor/team accounts.
- Device behavior is intentionally conceptual; low-level packet timing, packet loss, and numeric performance modeling are outside the scope of these tests.
- The platform is assumed to be available in a supported desktop browser with the latest stable release.

3. Referenced Main Use Cases

Use Case ID	Use Case Name	High-Level Description
UC-01	Complete Guided Tutorial Stage	Learner opens a lesson, reviews instructional content, completes the interactive task, receives feedback, and unlocks the next stage.
UC-02	Build Sandbox Topology	Learner uses the drag-and-drop canvas to place devices and connect them according to challenge constraints.
UC-03	Configure Network Devices	Learner edits device settings such as IP address, gateway pointer, DHCP setting, or wireless association.
UC-04	Validate Network Topology	System checks topology correctness against stage success criteria and returns green, amber, or red feedback.
UC-05	Run Request Simulation	Learner sends a simulated request and observes the request/response path through devices with explanatory tooltips.
UC-06	Save and Revisit Topology Portfolio	Learner saves a completed or custom topology, opens it later, and continues learning or sharing.
UC-07	Register, Authenticate, and Persist Progress	Learner creates/signs into an account and progress is saved across sessions.
UC-08	Monitor Cohort Progress	Instructor/admin reviews learner progress, completion, and challenge status for a classroom or team cohort.

4. Test Case Specification

TC-01 - Complete First Guided Tutorial Stage

Test Case ID	TC-01	Priority	High
Related Use Case(s)	UC-01, UC-04, UC-07	Test Level	System / UAT
Objective	Verify that a new learner can complete a guided tutorial stage, receive validation feedback, earn completion credit, and unlock the next stage.	Test Type	Functional
Preconditions	A learner account exists or guest mode is available; Arc 1 Stage 1 content is published; the learner has not previously completed Stage 1.	Test Data	Test learner: learner01@example.com; Stage: LAN basics / first tutorial stage.

Test Steps and Expected Results:

Step #	Action	Expected Result
1	Open NetPath and start the first guided tutorial stage.	Stage content loads without error.
2	Review the lesson explanation and demonstration content.	The interactive task is available after the explanation.
3	Complete the required interactive task using the provided starter topology.	Validation returns success for the correct solution.
4	Submit or trigger validation.	The learner receives a stage completion badge or completion indicator.
5	Continue to the next stage after success.	The next stage becomes unlocked and progress is saved.

Pass/Fail Criteria: The stage is marked complete, the next stage is available, and progress persists after page refresh or re-login.

TC-02 - Build a Valid Sandbox Topology

Test Case ID	TC-02	Priority	High
Related Use Case(s)	UC-02, UC-04	Test Level	System / UAT
Objective	Verify that a learner can build a challenge topology by dragging devices onto the canvas and connecting them according to stated constraints.	Test Type	Functional
Preconditions	Learner has access to a sandbox challenge with unlocked host, switch, router, and internet/ISP endpoint devices.	Test Data	Challenge: Connect multiple office hosts to the internet through the required network devices.

Test Steps and Expected Results:

Step #	Action	Expected Result
1	Open the assigned sandbox design challenge.	All unlocked devices are available in the palette.
2	Drag required devices from the device palette onto the canvas.	Dragged devices appear on the canvas and can be repositioned.
3	Draw wired links between ports to form the required topology.	Valid links are drawn between compatible device ports.
4	Observe real-time validation feedback while editing.	Invalid or incomplete topology produces amber/red feedback.
5	Complete the final topology that satisfies the challenge requirements.	Final correct topology produces green validation feedback.

Pass/Fail Criteria: The system accepts the completed topology and clearly identifies the design as valid.

TC-03 - Configure Device Network Settings

Test Case ID	TC-03	Priority	High
Related Use Case(s)	UC-03, UC-04	Test Level	System / UAT
Objective	Verify that device configuration settings can be edited and that validation responds correctly to both incorrect and corrected settings.	Test Type	Functional
Preconditions	Learner is in a stage that requires configuration of IP addresses,	Test Data	Device under test: router and host; invalid IP/gateway values

Test Case ID	TC-03	Priority	High
	gateway pointers, or DHCP settings.		followed by valid values.

Test Steps and Expected Results:

Step #	Action	Expected Result
1	Open a lesson or sandbox requiring device configuration.	Configuration panel opens for the selected device.
2	Select a host device and open the configuration panel.	The system accepts editable fields appropriate to the lesson stage.
3	Enter an invalid IP address or missing gateway and save.	Incorrect configuration causes red or amber validation feedback with a useful explanation.
4	Observe validation status.	Correct configuration updates the device state.
5	Correct the configuration values and save again.	Validation changes to green when all success criteria are satisfied.

Pass/Fail Criteria: The platform detects configuration errors, provides actionable feedback, and validates the corrected configuration.

TC-04 - Run Animated Request Simulation

Test Case ID	TC-04	Priority	High
Related Use Case(s)	UC-05, UC-04	Test Level	System / UAT
Objective	Verify that a learner can run a request simulation only after a valid topology exists and can observe the request and response path with explanatory tooltips.	Test Type	Functional
Preconditions	A topology has passed validation for the current challenge.	Test Data	Valid topology containing host, switch, router, DNS or internet destination as required by the stage.

Test Steps and Expected Results:

Step #	Action	Expected Result
1	Open a completed valid topology.	Send Request is enabled only when the topology is valid.
2	Confirm the validation indicator is green.	A visible animated marker travels along the expected device path.
3	Click Send Request.	Each device pause shows a tooltip explaining the forwarding decision.
4	Observe the animated request path across links and devices.	The response returns along the expected path or documented reverse path.
5	Observe the response path returning to the source.	Simulation completes without changing the saved topology configuration.

Pass/Fail Criteria: The learner can clearly follow the simulated request/response path from source to destination and back.

TC-05 - Save and Reopen Topology Portfolio Item

Test Case ID	TC-05	Priority	Medium
Related Use Case(s)	UC-06, UC-07	Test Level	System / UAT
Objective	Verify that a registered learner can save a sandbox topology and later reopen it from the topology portfolio.	Test Type	Functional
Preconditions	Learner is signed in; portfolio feature is enabled; learner has created a topology.	Test Data	Portfolio item name: Office Floor LAN v1.

Test Steps and Expected Results:

Step #	Action	Expected Result
1	Create or open a completed topology.	Save action prompts for or accepts a topology name.
2	Choose Save to Portfolio.	The topology is stored with device positions, connections, and configuration values.
3	Enter a portfolio item name and confirm save.	The portfolio lists the saved item for the signed-in learner.
4	Navigate away from the sandbox or sign out.	Opening the item restores the same topology state.

Step #	Action	Expected Result
5	Sign back in and open the topology portfolio.	The restored topology can be validated or edited.
6	Select the saved topology item.	

Pass/Fail Criteria: The saved topology reopens with the same canvas layout, devices, links, and settings that were originally saved.

TC-06 - Register and Persist Learner Progress Across Sessions

Test Case ID	TC-06	Priority	High
Related Use Case(s)	UC-07, UC-01	Test Level	System / UAT
Objective	Verify that learner account creation/sign-in persists completed stages and unlocked content across sessions.	Test Type	Functional
Preconditions	Authentication service is available; test learner email is not already registered or has a reset state.	Test Data	Test learner: learner02@example.com; completed stages: Stage 1 through Stage 3.

Test Steps and Expected Results:

Step #	Action	Expected Result
1	Register a new learner account or sign in with a test account.	Account registration/sign-in succeeds.
2	Complete multiple tutorial stages.	Completed stages are associated with the learner account.
3	Sign out or close the browser session.	Progress is written to persistent storage.
4	Sign in again using the same account.	After re-login, completed stages remain marked complete.
5	Review curriculum progress and unlocked stages.	Previously unlocked stages remain available.

Pass/Fail Criteria: Learner progress is consistent before and after sign-out/re-login.

TC-07 - Instructor Reviews Cohort Progress Dashboard

Test Case ID	TC-07	Priority	Medium
Related Use Case(s)	UC-08	Test Level	System / UAT
Objective	Verify that an instructor or team admin can view cohort progress, learner completion status, and challenge activity.	Test Type	Functional
Preconditions	Instructor/admin account exists; cohort contains multiple learners with varied completion states.	Test Data	Cohort: Intro Networking Spring 2026; Learners with completed, in-progress, and not-started stages.

Test Steps and Expected Results:

Step #	Action	Expected Result
1	Sign in as instructor/admin.	Only authorized instructor/admin users can access the dashboard.
2	Open the cohort or team dashboard.	Cohort summary metrics display accurately.
3	Select the target cohort.	Learner-level progress is visible.
4	Review completion metrics, learner list, and stage/challenge status.	Challenge/stage completion status reflects current learner activity.
5	Filter or sort the learner progress view if available.	Filtering/sorting does not change underlying progress data.

Pass/Fail Criteria: The instructor/admin can accurately determine cohort progress and identify learners who need support.

TC-08 - Undo, Redo, and Reset Sandbox Changes

Test Case ID	TC-08	Priority	Medium
Related Use Case(s)	UC-02, UC-03, UC-04	Test Level	System / UAT
Objective	Verify that learners can experiment safely by undoing, redoing, and resetting canvas actions without corrupting the topology state.	Test Type	Functional
Preconditions	Learner is in a sandbox challenge with at least two devices available.	Test Data	Canvas actions: add device, move device, connect devices, edit configuration.

Test Steps and Expected Results:

Step #	Action	Expected Result
1	Add devices to the canvas and create a link.	Undo reverses actions in the correct order.
2	Move a device and edit one configuration setting.	Redo reapplies undone actions in the correct order.
3	Use Undo repeatedly to reverse recent actions.	Validation feedback updates after each undo/redo action.
4	Use Redo to reapply actions.	Reset clears learner changes or restores the provided starter topology.
5	Use Reset to return the challenge to its initial state.	No duplicate, orphaned, or corrupted links remain after undo/redo/reset.

Pass/Fail Criteria: Canvas history operations preserve a consistent topology graph and correct validation state.

5. Traceability Matrix

The matrix below maps each test case to the main use cases it verifies. An "X" indicates direct coverage.

Test Case	UC-01	UC-02	UC-03	UC-04	UC-05	UC-06	UC-07	UC-08
TC-01 Complete First Guided Tutorial Stage	X			X			X	
TC-02 Build a Valid Sandbox Topology		X		X				
TC-03 Configure Device Network Settings			X	X				
TC-04 Run Animated Request Simulation				X	X			
TC-05 Save and Reopen Topology Portfolio Item						X	X	
TC-06 Register and Persist Learner Progress Across Sessions	X						X	
TC-07 Instructor Reviews Cohort Progress Dashboard								X
TC-08 Undo, Redo, and Reset Sandbox Changes		X	X	X				

6. Coverage Summary

Use Case ID	Use Case Name	Mapped Test Case(s)	Coverage Status
UC-01	Complete Guided Tutorial Stage	TC-01, TC-06	Covered
UC-02	Build Sandbox Topology	TC-02, TC-08	Covered
UC-03	Configure Network Devices	TC-03, TC-08	Covered
UC-04	Validate Network Topology	TC-01, TC-02, TC-03, TC-04, TC-08	Covered
UC-05	Run Request Simulation	TC-04	Covered
UC-06	Save and Revisit Topology Portfolio	TC-05	Covered
UC-07	Register, Authenticate, and Persist Progress	TC-01, TC-05, TC-06	Covered
UC-08	Monitor Cohort Progress	TC-07	Covered

7. Approval and Maintenance Notes

- The test suite should be updated whenever a new main use case, external integration, or platform tier is added.
- Detailed lower-level test cases should later be added for browser compatibility, accessibility, security, performance, and API contract behavior.
- Any failed test case should be linked to a defect record and retested after the defect is resolved.